THE
SCARA Robot

THE Series Safety Manual

(Compatible Controller: TSL3000/TSL3000E)

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Important

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

This manual is applicable to the following robot.
THE Series: THE400

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Introduction

Thank you very much for your purchase of SHIBAURA MACHINE's robot system.
This manual describes the safety measures for the system robot THE series built by SHIBAURA MACHINE.
Please read through this manual carefully and handle the robot according to the instructions given throughout the manual so that you can completely understand the performance of the robot and use its functions safety over the long years to come.

This manual has the following chapters.

- Use Range
  This chapter describes the use range intended by this product.

- Use Method
  This chapter describes main specifications and safe use method of this product.

- Warranty
  This chapter describes the scope of warranty.

- Standards, laws and regulations regarding safety of industrial robots
  This chapter describes main standards, laws and regulations that should be observed to use the robot.

- Safety Precautions
  This chapter describes important information on using the robot safely and properly.

- Service Network
  This chapter describes technical inquiry and service contact.

- Locations of Warning Labels
  This chapter describes the locations of warning labels affixed to the robot and controller.

- Safety Measures
  This chapter describes the safety functions of the robot and controller, and safety precautions for installing and operating the robot.

- Applicable Standards and Safety Measures
  This chapter describes the standards and safety measures with which the robot and controller are compliant.

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Use Range

This product is the horizontally articulated type industrial robot in which the manipulator has two horizontally rotating joints and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used, it will be guaranteed under customer's responsibility.

Use Method

The use conditions for the THE series robot are described. If the product is used in the state that differs from the conditions, unexpected risk may occur. Abide by the conditions for use.

1. Main Specifications of Product: See Specifications “1.4 Specifications Table” and Installation “3.1 Installation Environment” in the Robot Manual.
2. Restrictions on Use: Tables 1 and 2 give “Restricted Specification Sheet” indicating restrictions on use.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teaching pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint and the origin position. Work performed when the primary power is on: [1] Check the grease application state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teaching pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power to robot, controller, and optional parts while their covers are removed - Operation in other than the specified installation state (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for four (4) hours. - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis — —	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. — —
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: Maintenance: Disposal:	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m3 shall be secured. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. When entering a hazard zone, there must be a space that is not sandwiched between the robot and peripheral equipment. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. There must be a space in the hazard zone that is not sandwiched between the robot and peripheral equipment. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal.”

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	—
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operation procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	—
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power supply turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 5. Restricted Specifications Sheet

6. Other

Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.
We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period

The warranty period of the product is either of the following, whichever is earlier.

1) 18 months after delivery from our factory
2) 12 months after installation at your site

2. Warranty scope

1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
   Any repairing after the warranty period passes will be charged.

3. Disclaimer

The warranty shall not apply to the following cases.

a) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
b) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables[*1)].
c) Any failure regarding sensuous phenomena such as noise not affecting the functions
d) When any modification or disassembly without our consent is confirmed
e) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
f) Any failure or damage due to act of God, fires, and other force majeure
g) Internal data such as programs and points created or modified by users
h) When a product purchased in Japan is taken abroad

4. Precautions

1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
2) If the customer did not observe the “DANGER,” “WARNING,” and “CAUTION” described in this instruction manual, SHIBAURA MACHINE will not assume the responsibility for any consequential accident resulting in injury or death, damage or trouble.
3) “DANGER,” “WARNING,” “CAUTION” and other descriptions stipulated in this instruction manual are only those which can be assumed by SHIBAURA MACHINE without predicting all events in all situations. Please no that the range is limited.

(*1) Consumable parts refer to the replacement parts for maintenance as listed in instruction Manual “Robot Manual: Maintenance.”

Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy

Step 2: Safety protection - Installation of protective fences, etc.
Additional protection measures - Installation of emergency stop devices

Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, international standard ISO/IEC defines “Guide 51” that groups various safety standards in a hierarchical structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

- ISO 10218, ISO 10218-1

After risk assessment is carried out, provision of residual risk information to the user is standardized.

- IEC 82079-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

Safety Precautions

This manual contains important information for the safe use of the robot and the prevention of injury to the operator and others and the prevention of damage to property.
Read and understand the meanings of the following indications and symbols first, and observe these precautions.

Display of Signal Words and Their Meanings

Signal word and symbol	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1)] to the operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2)] and provides helpful advice about performing work.

1) Injury refer to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) Damage to property refers to extensive damage to property or materials.

Work Scope

The expected work scope for each life cycle is shown below.
In actuality, some tasks cannot be clearly classified and others are spread over multiple life cycles. These classifications are intended as a general guideline only, and so please check the safety information for all life cycles, and take protective measures before starting work.

- Transportation: From unloading packing boxes brought in by transportation equipment (truck, etc.) to carrying them to the unpacking location.

- Unpacking: From opening the packing box to taking out the product.

- Transfer: From taking out the product from the packing box to carrying it to the installation site.

- Installation: From installation of the product, wiring, piping, assembly of the hand and optional parts until it is powered on.

- Teaching: Changing parameters, teaching, programming, and checking operation.

- Operation: Operation of equipment incorporating the robot.

- Repacking: Shipping of a product to another installation site after performing operation verification.

- Maintenance: Maintenance and inspection of product.

- Disposal: Disposal of parts and other items that are no longer needed.

Transportation

This section provides important safety information for transporting the robot. Read this information together with " STEA1105:THE400 Robot Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- When lifting packing boxes, never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - The forklift must be operated by a qualified operator.
  - Align the center of the packing box with the center of the forks when lifting the packing box.
  - Do not allow people to approach the transportation route.

This section provides important safety information for transporting the controller. Read this information together with " STEA1107:TSL3000Controller Manual: Installation", and be sure to strictly observe the following items.

⚠ CAUTION

- Do not stack the packing boxes. The load could collapse, resulting in an injury.
- Do not allow the controller to get wet. Moisture could seep inside, resulting in a breakdown.
- Wear protective gloves and shoes when carrying the controller by hand. Dropping could result in crushed hands or feet.
- Use a cart or similar device for transporting the controller. Carrying by hand can lead to dropping, resulting in crushed hands or feet.

NOTICE

- Do not subject the controller to excessive vibrations during transportation. This could damage the controller.
- Do not transport upside down, and do not store on its side. This could damage the controller.

Unpacking

This section provides important safety information for unpacking the robot. Read this information together with " STEA1105:THE400 Robot Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- After the work is completed, be sure to always check that the robot is not damaged.
  Turning on the power for a malfunctioning robot can lead to an expected accident.
  - Check the appearance based on the external view shown in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or parts are missing, please contact the Shibaura Machine Sales Division.
- When carrying the packing boxes to the storage site, never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - Use a hand lifter or similar equipment.
  - Align the center of the packing box with the center of the forks when lifting the packing box.
  - Do not allow people to approach the transportation route.

⚠ WARNING

- When cutting the packing strap with the cutter, never point the blade edge towards a person.
  Otherwise, the person could get cut by the cutter.
  - Use a nippers or scissors.
  - Wear protective gloves.

⚠ CAUTION

- When storing the robot, be sure to insert in the packing box and secure it in place as it was delivered.
  Otherwise, the robot could fall over.
  - The customer must obtain new packing straps (PP bands).
  - If no packing box is available, use the mounting holes in the base to secure in place.

NOTICE

- After the work is completed, be sure to always check that the robot is not damaged.
  A damaged robot could malfunction.
  - Check the appearance based on the external view shown in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or parts are missing, please contact the Shibaura Machine Sales Division.
- Be sure to dispose of any unneeded packing materials in accordance with the customer's company regulations.
  When storing the packing materials, be sure to manage them so that they are not used for purposes other than packing the robot.
- When storing the robot, seal it in a plastic bag, and put a desiccant inside the bag.
  This prevents rust from forming and dust from adhering to the ball screw spline unit.
  - Before storing the robot, apply grease to the ball screw spline shaft.
  - For details on how to apply grease, see " STEA1105:THE400 Robot Manual: Maintenance".
  - Also apply grease to the ball screw spline shaft before use.
- Before operating a robot that was stored, be sure to perform breaking-in operation.
  During storage, the grease viscosity increases. If the robot is suddenly operated at a high speed, the poor lubrication could result in failure.
  - Be sure to perform operation at a low speed (about 20% override) for 5 minutes.
- Before operating a robot that was stored, be sure to replace the encoder batteries. See " STEA1105:THE400 Robot Manual: Maintenance".
  An alarm occurs before periodic inspection.
  - If power is not supplied to the robot for an extended period of time due to storage or other reasons, the data retention period by the encoder batteries will be shorter.
- When storing the robot, use the supplied fixtures to secure the arm and base, and then firmly secure the mounting sections. If placed directly on the floor, the robot will be unstable and will fall over.
- Never store the robot at a location exposed to direct sunlight, high temperature, or high humidity.
  This could cause the cable ducts, timing belt, and resin parts to deteriorate.
- Do not store the robot at a location where it is exposed directly to rain or water droplets, or at a location exposed to any toxic gas or liquid. This could cause premature deterioration of parts.

This section provides important safety information for unpacking the controller. Read this information together with " STEA1107:TSL3000Controller Manual: Installation", and be sure to strictly observe the following items.

⚠ CAUTION

- After unpacking, wear protective gear when disposing of unneeded wooden pallets and cardboard boxes. Otherwise, you could be cut by nails sticking out and by other sharp locations, resulting in an injury.
- Use a box cutter or similar tool when opening the packing box. You could cut your finger or hand if using a cutter with a sharp edge.
- Unpack at a stable location. If unpacked at an unstable location, the controller could fall and crush your hands or feet or strain your back.
- Wear protective gloves when unpacking. Your hand could get trapped between a cardboard box and the controller, resulting in an injury. You could also be injured by the corners of the controller or cardboard box.

NOTICE

- Do not allow the controller to get wet. Moisture could seep inside, resulting in a breakdown.

Transfer

This section provides important safety information for transferring the robot. Read this information together with " STEA1105:THE400 Robot Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- The robot should never be transferred by only one worker.
  The robot cannot be fully supported and could drop.
  - The robot weight is 15 kg.
  - Lift the robot using more than one person.
  - Use a cart or similar device to move to the transfer location.
  - Clear a transfer route, and make sure that a dropped robot would not hit anyone.
  - Wear protective gloves and safety shoes.
- When hoisting the robot, be sure to always observe the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the attached eyebolts.
  - The crane must be operated by a qualified operator.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
- Be sure to always hoist and lower the robot slowly.
  Otherwise, the robot could be tilted and fall.
  - Clear a transfer route, and make sure that no one goes under the suspended load.

⚠ WARNING

- When transferring the robot, be sure to always secure in place using the supplied fixing jigs.
  Otherwise, a worker could get caught in an arm that moved.
  - The weight of the arm section is approximately 9.5 kg.
  - Wear protective gloves to prevent slippage and to protect the hands.

NOTICE

- The robot should never be transferred by only one worker.
  The robot cannot be fully supported and could drop.
  - Do not use a product that has been dropped.
- Never touch the ball screw spline shaft with your bare hands.
  This could cause rust to form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to hoist and lower the robot gently.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- When lifting up the robot by workers, be sure to hold the appropriate sections shown in " STEA1105:THE400 Robot Manual: Installation".
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.
- After installation is completed, be sure to remove the fixing jig and eyebolts.
  If the robot is operated while the jig is mounted, the robot may be damaged or fail.

This section provides important safety information for transferring the controller. Read this information together with " STEA1107:TSL3000Controller Manual: Installation", and be sure to strictly observe the following items.

⚠ CAUTION

- Do not carry by holding the AC power supply unit. Carrying by holding the AC power supply unit could damage the controller or cause it to drop. Also, if AC power is being supplied, it could result in an electric shock.
- Wear protective gloves and shoes when carrying the controller by hand. Dropping could result in crushed fingers or feet.
- Use a cart or similar device for transporting and carrying. Carrying by hand can lead to dropping.
- Carry with the controller cables removed. Carrying with the cables connected could cause cable breakage, resulting in an electric shock.

NOTICE

- Do not subject the controller to excessive vibrations. This could damage the controller.
- Do not allow the controller to get wet. Moisture could seep inside, resulting in a breakdown.

Installation

This section provides important safety information for installing the robot. Read this information together with " STEA1105:THE400 Robot Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- Never turn on the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal movement in " STEA1105:THE400 Robot Manual: Installation".
- Never connect a power supply with a voltage exceeding the specified range to the robot hand wiring cable.
  This could cause the cable to overheat, resulting in a fire.
  - Before connecting, use a tester or similar device to check the voltage of the power supply.

⚠ DANGER

- Be sure to connect the cables to the controller after the robot is installed.
  Otherwise, a worker could receive an electric shock due to touching a cable that was broken due to a carried object that was dropped.
  - If a cable is broken, do not turn on the power.
- Be sure to always store the cables in ducts and similar protection.
  Otherwise, a worker could trip and fall over a cable on the floor.
  - Workers should wear a helmet to protect their head.
- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  An unexpected motion by the robot could damage the mechanical stopper and crash into someone.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.
- When fabricating the hand wiring cable, be sure to use the compatible wire.
  Otherwise, the cable could overheat or short circuit between the terminals, resulting in a fire.
- Be sure to lock the robot hand connector and secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the robot hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.

⚠ WARNING

- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- Be sure to use a robot hand having a structure so that it will not release the workpiece or other object even if the robot power or robot hand driving power supply is cut off.
  Otherwise, the workpiece or other object could fly off or drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - Conduct daily inspection on the air tube connected to the hand, and conduct periodic inspection on the air tubes in the robot, and replace them if they show any damage.
  - Install a fence or other barrier to protect from flying and dropped objects.
  - When using a robot hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply.

⚠ CAUTION

- When removing the fixing jig, never grasp the edges strongly.
  If excessive force is applied, a hand or finger could be cut on the fixing jig.
  - Wear protective gloves.
  - The bolts that are fastened to the robot may become hardened due to vibrations during transportation or transfer.
- Always check the working envelope before using the robot.
  If the working envelope is not limited as expected, you could be struck by an operating robot.

NOTICE

- Never turn on the robot in a location where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  This could cause a breakdown of the robot.
- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot could vibrate during operation. This could also lead to a malfunction of the robot.
  - Prepare the frame by referring to the load applied during horizontal movement in " STEA1105:THE400 Robot Manual: Installation".
- Never reuse a mechanical stopper that has collided with the arm and was used to change the work envelope.
  A damaged mechanical stopper will not function properly.
- Do not route piping together with the body harness.
  Doing so could apply excessive force on the body harness, resulting in breakage of the internal cables.
- Never use the robot under conditions outside the specification values.
  Otherwise, the performance and lifespan of the robot could be adversely affected.
- Never mistake the terminal arrangement on the hand wiring.
  Doing so could cause the connected parts to fail.
  - Before usage, use a tester or similar device to check the terminal arrangement.

NOTICE

- When operating the robot after an extended stop at a low temperature (10°C or less), be sure to perform operation at a low speed (about 20% of override) for 5 minutes.
  The grease viscosity increases in low-temperature environments. If the robot is suddenly operated at a high speed, the poor lubrication could result in failure.
  - Apply grease to the ball screw spline shaft before use.
- Be sure to route the cables after installation is completed.
  Otherwise, a carried object could drop on a cable that was already routed, resulting in breakage.
  - House the wiring in ducts or other protection.
- Be sure to install the robot on a level surface.
  Otherwise, the performance and lifespan of the robot could be adversely affected. This could also lead to a malfunction of the robot.
- Be sure to secure the frame where the robot is installed to a floor or wall.
  Otherwise, the frame will move due to the force applied during robot operation.
- Be sure to always connect a ground wire.
  Otherwise, the robot may fail due to noise.
- When designing, manufacturing, and mounting the mechanical stoppers, be sure to observe the following points.
  - Be sure to set the changed work envelope to within the range shown in the specification table and external view drawing.
    The robot cannot be operated in a work envelope that is outside the specifications.
  - Set the origin position information again by following the procedure in " 8.9 ③Other Zero Setting ".
    If a mechanical stopper is changed, the function described in " 8.9.1 Restoring Zero Position Data by HOME Function [4] How to Restore Data by HOME3 or HOME4" is disabled. At installation, enter the origin position data by following the procedure in "8.9.1 Restoring Zero Position Data by HOME Function [2] Setting HOME1 and HOME2".
  - A function is also available to restore the origin position using a setting jig provided by the customer. If using the procedure in "8.9.1 Restoring Zero Position Data by HOME Function [3] How to Restore Data by HOME1 or HOME2", enter the origin position data using the procedure in "8.9.1 Restoring Zero Position Data by HOME Function [2] Setting HOME1 and HOME2" during the installation process.
  - A robot failure caused by a mechanical stopper is outside the scope of the warranty.
- Be sure to set the software limits inside the mechanical stoppers.
  If the software limits are not set inside the mechanical stoppers, the robot will collide with the mechanical stoppers during operation.
- If the service tap is used to mount parts, be sure to always check whether the parts will interfere with the robot or peripheral equipment during operation. Otherwise, this could result in a breakdown of the robot or peripheral equipment.

NOTICE

- Be sure to use the PAYLOAD command.
  Doing so could adversely affect the performance and lifespan of the robot.
- Even if the load center of gravity offset value is 100 mm or less, on robot hands with a long projection length from the 4th axis rotation center, micro-vibrations at the axis center can be amplified to cause large vibrations on the hand and workpiece tip.
- Be sure to always use the PAYLOAD command.
  Otherwise, vibration or overrun could reduce the lifespan of the robot or damage the robot.
  - The moment of load inertia should be used within the tolerance listed in "STEA1213: THE400 TS5000 Series Robot Manual: Installation".
  - Even if there is no load center of gravity offset, when the moment of inertia is large, the robot could vibrate. In these cases, calculate the virtual center of gravity offset L (mm) from the moment of inertia ($kg \cdot m^2$) and weight M (kg).
    $L=\sqrt{(J\times10^6/M)}$
    Enter this calculation result.
    PAYLOAD={M,L}
  - Micro-vibrations could occur depending on the robot orientation.
  - If the vibrations or overrun does not subside, use together with the SPEED command or DECEL command.
- Even if the weight is within the maximum payload weight, vibrations could occur depending on the robot orientation, such as when a heavy object is held and the 3rd axis is lowered. In these cases, use the SPEED command, ACCEL/DECEL command, LOCUS command, or similar commands to adjust the operating speed and acceleration/deceleration so that vibration does not occur.
  For the SPEED command, ACCEL/DECEL command, and LOCUS command, refer to “3 Explanation of Command Words” in the separate instruction manual “Robot Language Manual”.
- If the robot vibrates when using manual guidance, be sure to set the PAYLOAD in manual mode by referring to “10.16 Payload Settings During Manual Guidance” in the separate instruction manual “STE80720:INSTRUCTION MANUAL”.
  Otherwise, vibration could reduce the performance and lifespan of the robot.
  - If the load weight or center of gravity offset is large, the robot will vibrate.
- When fabricating the hand wiring cable, be sure to use the compatible wire.
  Otherwise, overheating or short circuit could occur, resulting in burning of the cable.
- Be sure to use a highly flexible and twisting-resistant robot cable for the hand wiring cable.
  If a general-purpose cable is used, the wire may break during robot operation.
  - Secure the cable to the arm using a saddle or other mechanism to prevent the cable from being swung around.
- Never connect the hand wiring cable to a power supply whose voltage exceeds the specified range.
  This could cause the cable to overheat, resulting in a fire.
  - Before connecting, use a tester or similar device to check the voltage of the power supply.

NOTICE

- Be sure to perform hand wiring and piping while checking the following status during robot operation. Otherwise, the robot and hand will break down prematurely.
  - Check whether a load is being applied to the hand connector.
  - Check whether the hand wiring and piping are contacting the robot or peripheral equipment and are worn.
- The cable between the robot and controller is not a movable cable. If the robot is mounted on a traveling axis or similar structure, the cable will be damaged prematurely.
- The solenoid valve air tubes are not supplied with the robot.
  They should be provided by the customer.

This section provides important safety information for installing the controller. Read this information together with " STEA1107:TSL3000Controller Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- Never install in locations with water exposure that exceeds the protection rating. Doing so could lead to an electric shock or fire.
- Never install near flammable objects. If sparks occur due to a breakdown, nearby flammable objects could ignite, resulting in a fire.
- Do not place objects on top of the controller. (Stacking is also prohibited.)
  - Lack of ventilation space and heat buildup could cause a fire.
  - Because the design assumes that no objects are placed on top of the controller, the controller could collapse due to being unable to support the object, resulting in a breakdown of the controller or crush someone's hands or feet.
- Never use unauthorized cables. Fabricating and use of unauthorized cables can result in a fire.
- Do not connect to unauthorized connectors or terminals. Unauthorized connectors and terminal wires could result in a breakdown of the controller, electric shock, or a fire.
- Do not insert conductive objects or your fingers inside the power supply terminals. An electric shock could occur.
- Be sure to always install the controller outside of the robot's hazard zone where you can oversee the robot's motion.
  If the controller is operated in a hazard zone or in a location where the robot or peripheral equipment is not visible, an operator or worker in the hazard zone will be struck by the robot or peripheral equipment.
- If the controller must be installed inside the hazard zone for unavoidable reasons, put the teach pendant outside the hazard zone, and set up so that the controller's system signal is used to output the robot status externally to a signal light or LED for displaying the status.
  If there is no means for checking the robot status, a worker who, due to unavoidable reasons, must check the controller installed in the hazard zone will be struck by the robot or peripheral equipment.
- When installing the controller in a closed space, such in a control panel, use air vent holes, forced ventilation, and indirect cooling to prevent heat buildup. If the controller is installed in a closed space, heat will be trapped, resulting in a breakdown or fire.
- Install the controller inside the control panel or in a location where sufficient distance can be provided so that it cannot come into contact with the operator. If the controller is installed within the operator's range of motion, the operator could fall over with arms or legs caught in a cable, or could receive an electric shock due contact with the energized parts.
- Do not install on an unstable location. The controller could fall and crush someone's hands or feet.
- Be sure to always perform the wiring for the robot after installation is completed.
  - If the controller is installed or transported while the wiring is connected, a cable could be broken, resulting in an electric shock.
  - Also, your legs could get caught in a cable, causing you to trip and fall.
- Provide a dedicated duct or similar protection to protect the cable lead-in of the control panel. If the cable is routed without using a dedicated duct, the cable could break, causing a fire or electric shock. The cable could break, causing a fire or electric shock.
- Mount a cover or other protection for the buttons and switches of the control unit connected to the controller. Accidental contact could turn on a signal, causing the robot to move, resulting in a collision.

⚠ DANGER

- Display the execution mode of the respective controller on the operation device. If the operator does not know the execution mode, a collision could occur with the robot due to an unexpected movement.
- Be sure to always connect a ground wire. Usage without connecting the ground wire can result in ground leakage, electric shock, or a fire.
- After wiring, use a tester or other device to check the voltage and for any short-circuits. If the power is turned on while the wiring is incorrect or while the controller is malfunctioning, an electric shock or fire could occur.
- Be sure to connect all connectors securely. If the connector is not engaged properly, static electricity could enter the controller and cause it to malfunction or cause a collision with the robot.

⚠ WARNING

- Turn off the power before connecting and disconnecting cables from the controller. Touching the terminal of a disconnected cable while the power is on could result in an electric shock, and connecting and disconnecting the cable while the power is on could damage the ICs in the board.
- When disconnecting the cable, pull it out by grasping the connector while holding down the controller.
  - Pulling the cable itself could break the cable and result in an electric shock.
  - The controller could fall over, resulting in a breakdown.

NOTICE

- Never supply a current exceeding the rated output. Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts, and could result in a fire.
- If the LAN, OUTPUT, and SD connectors will not be used, be sure to attach the connector caps (dummy SD card for the SD connector). If these connectors do not have any protection, electrostatic charges could cause a breakdown.
- Provide ventilation space around the controller of at least 50 mm on the right, left, and rear sides, 100 mm on the top side, and the height of the rubber feet (11 mm) on the bottom side. If no ventilation space is provided, the controller could overheat, resulting in a breakdown.
- When installing the controller in a closed space, such in a control panel, use air vent holes, forced ventilation, and indirect cooling to prevent heat buildup. If the controller is installed in a closed space, heat will be trapped, resulting in a breakdown or fire.
- If the controller will be installed in the control panel, use fixing jigs or other means to secure the controller in place. If the control panel vibrates, its inner components could move, and the controller could drop, resulting in a fire.
- If the controller will be installed in the control panel, use fixing jigs or other means to secure the controller in place. If the control panel vibrates, its inner components could move, and the controller could drop, resulting in a fire.
- Set the robot hand output signal to 0.1 A per output (max. 0.8 A for total 8 outputs). Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts, and could result in a fire.
- Apply a voltage in a range of 24V±10% to the input/output external wire. If a voltage exceeding the tolerance is applied, the controller will be damaged, and a fire could result.

Teaching

This section provides important safety information for teaching. Read this information together with " STEA1105:THE400 Robot Manual: Installation", and be sure to strictly observe the following items.

⚠ DANGER

- Never turn on the power for the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  - Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal movement (Table 3.2).
- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  An unexpected motion by the robot could damage the mechanical stopper and crash into someone.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.
- Be sure to lock the robot hand connector and secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the robot hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.
- When performing the teaching operation in the dangerous area, the operator must carry the controller's master key switch and the teaching pendant in order to prevent a third person from operating the machine.
  Otherwise, you could be struck by a robot that has been started by someone else.
  - Post a notice near the controller that work is in progress.
  - Assign a work supervisor who is stationed outside the hazard zone of the robot.
  - The supervisor should ensure the safety of the worker and any other people approaching the robot hazard zone.
  - Make sure that the worker and the supervisor receive safety training on tasks related to the inspection of industrial robots as specified by the laws and regulations of individual countries.
- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ WARNING

- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- Be sure to use a robot hand having a structure so that it will not release the workpiece or other object even if the robot power or robot hand driving power supply is cut off.
  Otherwise, the workpiece or other object could fly off or drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - Conduct daily inspection on the air tube connected to the hand, and conduct periodic inspection on the air tubes in the robot, and replace them if they show any damage.
  - Install a fence or other barrier to protect from flying and dropped objects.
  - When using a robot hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply.
- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50℃.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- Always check the working envelope before using the robot.
  If the working envelope is not limited as expected, you could be struck by an operating robot.
- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

NOTICE

- Never touch the ball screw spline shaft with your bare hands.
  This could cause rust to form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could reduce the performance and lifespan of the robot. This could also lead to a malfunction of the robot.
- When pressing the robot by hand to move it, be sure to hold the gray sections shown in Figure 9.1.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.

This section provides important safety information for teaching. Read this information together with " STEA1107:TSL3000Controller Manual ", and be sure to strictly observe the following items.

⚠ DANGER

- Be sure to always perform operations as described in the instruction manual. Otherwise, a collision could occur with the robot, or an electric shock could occur.
- Be sure to perform teaching outside the hazard zone. Otherwise, a collision could occur with the robot. If teaching must be performed within the hazard zone due to unavoidable reasons, be sure follow the guidelines below.

(1) Always perform the work using two people. One person performs the work, and the other person monitors from outside the hazard zone. Each person works to prevent mistaken operation.

(2) When teaching in the hazard zone, the operator should hold the master key of the controller and the teach pendant to prevent operation by a third party.

(3) The operator should work in a position where the emergency stop button can be pressed at any time, and should also work in a location where, after fully checking the robot's work envelope and surrounding shielding, he or she can immediately evacuate in case of an abnormality.

(4) The supervisor monitors the work from a location where he or she can oversee the entire robot and immediately presses the emergency stop button in case of an abnormality. Other people are kept away from the hazard zone. Failure of the operator or others to follow the instructions of the supervisor could result in an accident.

- Be sure to turn off the power if an abnormality occurs when the equipment is turned on or if the controller's "READY" LED does not light up. Otherwise, an electric shock or fire could occur.
- Be sure to always perform daily inspection before operating the equipment.
  Failure to perform daily inspection could lead to an accident.
- When the servo is turned on, be sure to always be outside the hazard zone.
  Otherwise, a collision could occur due to an unexpected movement by the robot.
- In case of malfunction or failure, apply an emergency stop to the robot and turn off the controller.
  Otherwise, a fire or collision with the robot could occur.

⚠ WARNING

- Never enter the hazard zone during test operation. Otherwise, a collision could occur with the robot.
- Never leave any equipment in the work space that could interfere with your work.
  This could cause you to trip and fall. If you concentrate on teaching, you could fail to pay attention to your surroundings and collide with peripheral devices.
- Do not allow people to approach the equipment other than the specified operators. Otherwise, a collision could occur with the robot.
- Never open the cover of the controller during operation. Do not connect or disconnect connectors. An electric shock could occur. A collision could occur due to an unexpected movement by the robot.
- Never change the data in the system configuration files during operation. This could cause the robot to perform an abnormal movement, resulting in a collision.

⚠ WARNING

- Be sure to stop using the robot if any abnormal operation is observed when performing robot guidance. Otherwise, a serious accident could occur.
- Stop operation if there is any unusual noise or odor from the controller or robot, or if it behaves differently than expected. Continued operation while there is an abnormality could lead to a fire or a collision due to an unexpected movement.
- When teaching in a hazard zone, work with the teach pendant or with the master key in your possession. If another person starts operating the robot while you are working, a collision with the robot could occur.
- Turn off the servo when leaving the work area. Otherwise, another worker could mistakenly assume that the robot is stopped, resulting in a collision with the robot.
- In case of malfunction or failure, apply an emergency stop to the robot and turn off the controller. Otherwise, a fire or collision with the robot could occur.

⚠ CAUTION

- All hand output signals are in an OFF state when the controller power is shut off and turned on. Design the end effector so that objects will not drop from the hand due to a power outage or similar situation. Objects dropped from the end effector could crush the hands or feet of the operator.

NOTICE

- Be sure to turn off the servo while the robot is stopped before turning off the main power. Otherwise, if the main power is turned off while the robot is operating, the arm will run idly and collide with peripheral devices.
- Always check the execution status of the program when performing automatic operation. Failure to perform a check could result in interference with the peripheral devices and a breakdown of the equipment.

Operation

This section provides important safety information for teaching. Read this information together with "STEA1107:TSL3000Controller Manual: Installation", and be sure to strictly observe the following items.

⚠DANGER

- Never turn on the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal movement (Table 3.2).
- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  An unexpected motion by the robot could damage the mechanical stopper and crash into someone.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.
- Always check the working envelope before using the robot.
  If the working envelope is not limited as expected, you could be struck by an operating robot.
- Be sure to lock the robot hand connector and secure the cable to the arm.
  If a load is applied to the connector and it is disconnected, the hand could move unpredictably.
  - Adjust the robot hand wiring and piping while checking the status during robot operation to make sure that they are not in contact with the robot or peripheral equipment and that no load is being applied to the connectors.
- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ DANGER

- When performing the teaching operation in the dangerous area, the operator must carry the controller's master key switch and the teaching pendant in order to prevent a third person from operating the machine.
  Otherwise, you could be struck by a robot that has been started by someone else.
  - If there are no impediments to rescue, apply an emergency stop.

⚠ WARNING

- Be sure to use a robot hand having a structure so that it will not release the workpiece or other object even if the robot power or robot hand driving power supply is cut off.
  Otherwise, the workpiece or other object could fly off or drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - Conduct daily inspection on the air tube connected to the hand, and conduct periodic inspection on the air tubes in the robot, and replace them if they show any damage.
  - Install a fence or other barrier to protect from flying and dropped objects.
  - When using a hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply until a safety check is completed.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50℃.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

This section provides important safety information for operation. Read this information together with " STEA1107:TSL3000Controller Manual ", and be sure to strictly observe the following items.

⚠ DANGER

- Never operate the equipment when it is damaged or has missing parts.
  Doing so could lead to an electric shock or fire.
- Do not insert conductive objects or your fingers inside the power supply terminals. An electric shock could occur.
- Display the operation mode of the respective controller on the operation device. If the operator does not know the operation mode, a collision could occur with the robot due to an unexpected movement.
- Be sure to always perform operations as described in the instruction manual.
  Otherwise, a collision could occur with the robot, or an electric shock could occur.
- Be sure to turn off the power if an abnormality occurs when the equipment is turned on or if the controller's "READY" LED does not light up.
  Otherwise, an electric shock or fire could occur.
- When the servo is turned on, be sure to always be outside the hazard zone.
  Otherwise, a collision could occur due to an unexpected movement by the robot.
- In case of malfunction or failure, apply an emergency stop to the robot and turn off the controller.
  Otherwise, a fire or collision with the robot could occur.

⚠ WARNING

- Never open the cover of the controller during operation. An electric shock could occur.
- Never change the data in the system configuration files during operation. This could cause the robot to perform an abnormal movement, resulting in a collision.
- If there are multiple controllers in the equipment, ensure that they operate the robot using the same operation mode. If the operation modes are different, a collision could occur with the robot due to an unexpected movement.
- Never enter the hazard zone during test operation. A collision could occur with the robot.
- Do not allow people to approach the equipment other than the specified operators. A collision could occur with the robot.
- Be sure to always perform daily inspection before operating the equipment. Failure to perform daily inspection could lead to an accident.
- Stop operation if there is any unusual noise or odor from the controller or robot, or if it behaves differently than expected. Continued operation while there is an abnormality could lead to a fire or a collision due to an unexpected movement.
- Never enter the hazard zone of the robot during automatic operation. If you enter the hazard zone during operation, the robot could collide with you.
- Be sure to check operation of the safety protection devices before starting automatic operation. If the safety protection devices have failed, you will be unable to stop the equipment in the event of an emergency, and a collision with the robot could occur.

NOTICE

- Do not places any obstacles within the robot's work envelope. This could cause a collision with the robot.
- Set the robot hand output signal to 0.1 A per output (max. 0.8 A for total 8 outputs). Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts, and could result in a fire.
- Supply a voltage in a range of 24V±10% to the external input/output power supply (24V DC). If a voltage exceeding the tolerance is applied, the controller will be damaged, and a fire could result.
- Be sure to turn off the servo while the robot is stopped before turning off the main power.
  Otherwise, if the main power is turned off while the robot is operating, the arm will run empty and collide with peripheral devices.
- Always check the execution status of the program when performing automatic operation.
  Failure to perform a check could result in interference with the peripheral devices and a breakdown of the equipment.

Repacking

This section provides important safety information for repacking. Read this information together with " STEA1107:TSL3000Controller Manual ", and be sure to strictly observe the following items.

⚠DANGER

- Never turn on the power for the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  - Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- The robot should never be transferred by only one worker.
  The robot cannot be fully supported and could drop.
  - The robot weight is 15 kg.
  - Lift the robot using more than one person.
  - Use a cart or similar device to move to the transfer location.
  - Clear a transfer route, and make sure that a dropped robot would not hit anyone.
  - Wear protective gloves and safety shoes.
- Never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - Use a hand lifter or similar equipment.
  - Align the center of the packing box with the center of the forks when lifting the packing box.
  - Do not allow people to approach the transfer route.
- When lifting packing boxes, never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - The forklift must be operated by a qualified operator.
  - Align the center of the packing box with the center of the forks when lifting the packing box.
  - Do not allow people to approach the transportation route.

⚠DANGER

- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  An unexpected motion by the robot could damage the mechanical stopper and crash into someone.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.
- Always check the operation status before approaching a stopped robot.
  Otherwise, if the robot was operating, a collision could occur when the robot restarts its motion.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- When performing the teaching operation in the dangerous area, the operator must carry the controller's master key switch and the teaching pendant in order to prevent a third person from operating the machine.
  Otherwise, you could be struck by a robot that has been started by someone else.
  - Post a notice near the controller that work is in progress.
  Assign a work supervisor who is stationed outside the hazard zone of the robot.
  - The supervisor should ensure the safety of the worker and any other people approaching the robot hazard zone.
- Do not remove the robot until the power is turned off and the cable is removed from the controller.
  Otherwise, a worker could receive an electric shock due to touching a cable that was broken when the robot is dropped.
- When hoisting the robot, be sure to always observe the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the attached eyebolts.
  - The crane must be operated by a qualified operator.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
- Be sure to always hoist and lower the robot slowly.
  Otherwise, the robot could be tilted and fall.
  - Clear a transfer route, and make sure that no one goes under the suspended load.

⚠ WARNING

- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.
- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
  - After turning off the power, remove the ground wire.
- Be sure to check that the robot is using a robot hand having a structure so that it will not release the workpiece or other object before turning off the robot power or robot hand driving power supply.
  Otherwise, the workpiece or other object could drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot hand signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - When using a hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply until a safety check is completed.
- When transferring the robot, be sure to always secure in place using the supplied fixing jigs.
  Otherwise, a worker could get caught in an arm that moved.
  - The weight of the arm section is approximately 9.5 kg.
  - Wear protective gloves to prevent slippage and to protect the hands.
- Never use an awkward posture when inserting the robot into the packing box.
  Overexertion could injure the worker's back.
  - Provide space where work can be performed easily.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50℃.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.
- Always check the working envelope before using the robot.
  If the working envelope is not limited as expected, you could be struck by an operating robot.
- Be sure to secure the robot to be re-transported in a packing box as it was delivered.
  Otherwise, the robot could fall over.
  - The customer must obtain new packing straps (PP bands).
  - If no packing box is available, use the mounting holes in the base to secure in place.

NOTICE

- Never touch the ball screw spline shaft with your bare hands.
  This could cause rust to form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- The robot should never be transferred by only one worker.
  The robot cannot be fully supported and could drop.
  - Do not use a product that has been dropped.
- Be sure to hoist and lower the robot gently.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- Be sure to secure the robot to be shipped in a packing box as it was delivered, and then remove the eyebolts.
  If the robot is moved while the eyebolts are mounted, the robot could be damaged or fail.
  - This is a mechanism indicating that a robot with the eyebolts mounted cannot be operated.
  - Keep the removed eyebolts with the robot.
- When lifting up the robot by workers, be sure to hold the gray portions shown in Figure 2.3.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.

Maintenance

This section provides important safety information for maintaining the robot. Read this information together with " STEA1105:THE400 Robot Manual: Maintenance", and be sure to strictly observe the following items.

⚠ DANGER

- Never turn on the power for the robot in an environment where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Doing so could result an electric shock due to a leakage current.
- Never turn on the power for the robot in an environment with dust or flammable gas.
  Doing so could lead to a fire.
  Do not place flammable materials near the robot because they could ignite due to a spark if a malfunction occurs.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a malfunction due to noise could cause an expected accident.
- Never touch fluid that has leaked from the encoder batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Always check the operation status before approaching the robot.
  Otherwise, a collision could occur with the robot when it starts to move.
  - Check the operation status displayed by the controller indicator lamps and the teach pendant.
  - The operation status of the robot is output as signals (servo operation ready, external mode in progress, operation ready, automatic operation in progress, fault occurred, and emergency stop in progress). Connect to the display unit or similar device.
- When performing the teaching operation in the dangerous area, the operator must carry the controller's master key switch and the teaching pendant in order to prevent a third person from operating the machine.
  Otherwise, you could be struck by a robot that has been started by someone else.
  - Post a notice near the controller that work is in progress.
  - Assign a work supervisor who is stationed outside the hazard zone of the robot.
  - The supervisor should ensure the safety of the worker and any other people approaching the robot hazard zone.
  - If the teach pendant is not used, carry an emergency stop device with you when performing work.
  - If the power will be turned on, make sure that the worker and the supervisor receive safety training on tasks related to the inspection of industrial robots as specified by the laws and regulations of the respective country.
- Before turning on the power, always make sure that no one is inside the robot's hazard zone.
  A collision could occur due to an unexpected movement by the robot.
  - Make sure that the operation area limited by the mechanical stoppers is also clear of people. Use of the mechanical stoppers does not guarantee stopping of the robot.

⚠ DANGER

- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.
- Never modify parts except those that are expressly permitted.
  Otherwise, the robot could fail, and cause an expected accident.
- Be sure to always check that the polarity of the encoder batteries is correct.
  If the polarity is wrong, electrolyte could leak from the encoder batteries.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.
- Be sure to replace parts with the Shibaura Machine-specified service replacement parts.
  Use of unauthorized parts could lead to an unpredictable accident.
  - All items other than the encoder batteries are special-specification products. To order them, please contact Shibaura Machine.
  - Be sure to contact the Shibaura Machine servicing company to request any replacement work other than replacement of the encoder batteries. A fee will be charged for any replacement work.
  - Shibaura Machine does not cover any accidents or damage resulting from replacement work performed by the customer or its results.

⚠ WARNING

- Be sure to check that the robot is using a robot hand having a structure so that it will not release the workpiece or other object even if the robot power or robot hand driving power supply is cut off.
  Otherwise, the workpiece or other object could drop.
  - When the robot is turned off, the signal output to the hand is also turned off.
  - The robot output signals (DOUT201 to DOUT208) are not reset by performing an operation to reset all output signals (such as the system input signal "Output signal reset" or RESET DOUT command).
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other barrier to protect from dropped and flying objects.
  - When using a hand that releases the workpiece when the robot power or hand driving power supply is turned off, post notices to not turn off the power or driving power supply until a safety check is completed.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.
- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.
- When releasing the motor brake, be sure to always support the ball screw spline shaft.
  Otherwise, the shaft will drop due to its own weight as soon as the brake is released.
  - When supporting the shaft, be careful that your hand and fingers do not get pinched in between.
  - If you notice anything unusual during the work, immediately let go of the brake release switch. The brake will be also applied by turning off the power to the controller.
  - Do not mistakenly press the brake release switch on the robot 2nd arm top surface.

⚠ CAUTION

- Always check the working envelope before operating the robot.
  If the working envelope is not limited as expected, you could be struck by a moving robot.
- When rotating the 4th axis by hand, never insert your hand or fingers between the upper or lower mechanical stopper and the robot or under the robot hand.
  Your hand or fingers could get caught in the mechanical stopper or robot hand.
  - When the 4th axis rotates, the ball screw spline shaft moves up and down.
  - The 4th axis motor is not equipped with a brake. If the robot hand is touched by hand while the servo power is off, the 4th axis rotates.

NOTICE

- Be sure to conduct periodic maintenance and inspection.
  Failure to perform maintenance and inspection could reduce the performance and lifespan of the robot. This could also lead to a malfunction of the robot.
- Never modify parts except those that are expressly permitted.
  Otherwise, the performance and lifespan of the robot could be adversely affected. This could also lead to a malfunction of the robot.
- Be sure to always perform daily inspection for the air tube.
  Replace the air tube immediately if it is damaged.
- Never touch the ball screw spline unit with your bare hands.
  Touching it with bare hands causes the ball screw spline to rust.
  - Be sure to always wear gloves.
- When greasing and applying corrosion inhibitor, be sure to cover the peripheral equipment.
  Grease and corrosion inhibitor could drip on the peripheral equipment.
  - Wipe off any excess grease and corrosion inhibitor with a waste cloth.
- Be sure to check the grease application state daily.
  Insufficient grease or rust formation could cause scratches on the sliding surface, resulting in reduced performance and lifespan of the robot. This could also lead to a malfunction of the robot.
- Be sure to use the specified types of grease.
  Otherwise, the performance and lifespan of the robot could be adversely affected. This could also lead to a malfunction of the robot.
- If the origin is set again using the ZEROP function or REORG function, be sure to perform teaching of the teaching points again.
  The zero position shifts slightly due to error when setting. If the origin position is used without setting it again, the robot could collide with the peripheral equipment.
- Be sure to always replace the encoder batteries during the annual inspection.
  If the position detector data is lost, the robot can no longer be operated.
  - If the encoder battery voltage drops even further, a “Battery alarm” will occur. At this stage, data is not lost yet. If the encoder batteries are replaced, the alarm is automatically cleared.
  - If the battery voltage drops even further, a “Battery alarm” will occur. At this stage, the data is lost. A position detector error alarm will also occur.
  - If the data is lost, the origin position must be set again. For details on setting the origin position, refer to "14. Robot Origin Position and Position Detector Error".

NOTICE

- When the robot is not used while the power turned off, be sure to replace the encoder batteries every two months.
  Because the encoder batteries use a large amount of power while the power is turned off, the position detector data will be lost before the periodic inspection is conducted.
- To set the origin position again using the ZEROP function or REORG function, perform the following operation procedure.
  Because the 3rd axis operates in tandem with the 4th axis, the origin position is not restored if the steps are performed in the wrong order.
  Operation procedure
  1) Move the robot to the specified position.
  2) Move the cursor to E4 on the screen, enter the data, and restore the origin position of the 4th axis.
  3) Move the cursor to E3 on the screen, enter the data, and restore the origin position of the 3rd axis.
  - E1 (1st axis) and E2 (2nd axis) do not have any specific order for setting their origin position again.
- Be sure to enter data to HOME1 and HOME2 for the REORG function when installing the robot.
  Even if the data is entered after the origin position has shifted, it cannot be used as data for restoring the origin position.
- Be sure to replace parts with the Shibaura Machine-specified service replacement parts.
  Use of unauthorized parts could lead to reduced robot performance and lifespan. This could also lead to a malfunction of the robot.

This section provides important safety information for maintaining the controller. Read this information together with " STEA1107:TSL3000Controller Manual: Maintenance", and be sure to strictly observe the following items.

⚠ DANGER

- Never perform any parts replacement or modification except those described in the instruction manual. Reduced performance or malfunctions could result in electric shock, fire, or collision due to unexpected robot movements.
- When opening the cover of the controller, do not insert liquids such as water or foreign objects inside. This could cause a short-circuit inside the controller, resulting in an electric shock or fire.
- Do not incinerate, disassemble, short-circuit, or charge the batteries. This could rupture, resulting in a fire. Dispose of the batteries in accordance with your company's regulations.
- Shibaura Machine special parts must be used for some service replacement parts. Unauthorized connectors and terminal wires could result in a breakdown of the controller, electric shock, or a fire.
- Be sure to always request part replacement from the Shibaura Machine servicing company for parts replacement that is not described in this manual. (A fee will be charged for this work.) Shibaura Machine will not be liable for any accidents or damage that arises due to replacement work performed by the customer or its results.
- Install the switching power supply as described in the instruction manual. Installing the unit upside-down or with the right and left sides reversed, or using a different connector could cause a short-circuit or disconnection, which could result in an electric shock or fire.
- Be sure to replace parts with the Shibaura Machine-specified service replacement parts. Use of unauthorized parts could lead to an unpredictable accident.
- All items other than the position detector batteries are special-specification products. To purchase and order them, please contact Shibaura Machine.
- Except when replacing button batteries and fuses, be sure to always request replacement work from the Shibaura Machine servicing company. A fee will be charged for this work. Shibaura Machine will not be liable for any accidents or damage that arises due to replacement work performed by the customer or its results.

⚠ WARNING

- Wait about two minutes after disconnecting the power supply before touching the terminals of the controller or the internal circuit board. Residual voltage may remain in the high-capacity capacitors in the controller, which could result in an electric shock.
- Use Shibaura Machine-authorized spare parts when replacing parts. Use of unauthorized parts could cause a fire or a collision due to an unexpected robot movement.
- Be sure to wear protective gear and remove the power plug before starting work. If the power is turned on, the power supply board, servo board, switching power supply, and other components will be electrically charged, and could cause an electric shock.
- Turn off the power before opening the cover of the controller. If the cover is opened while power is being supplied, an electric shock could result.
- Perform the following actions whenever entering the movable range of the robot.
  - Wear protective gear
  - Place the teach pendant in a location accessible to the worker
  - Change to TEACH mode (remove the mode selector key so that the mode cannot be changed)

⇒ If the above actions are not taken, an emergency stop could be cleared by a third party during daily inspection work within the moving range, and that person could be struck by a robot running under continuous operation.

- Use the voltage measurement and LEDs to check that a PN voltage is not remaining in the capacitor. An electric shock will occur if a PN voltage remains.
- Do not use a metal scale while the power is being supplied. An electric shock will occur if contacting the energized parts.

NOTICE

- Be sure to replace parts with the Shibaura Machine-specified service replacement parts. Use of unauthorized parts could lead to reduced robot performance and lifespan. This could also lead to a malfunction of the robot.
- All files in the controller will be deleted by USER BACKUP LOAD. Before executing, use the file transfer or other function to save the files in the controller to an external device.

Disposal

This section provides important safety information for disposal of the robot. Read this information together with " STEA1105:THE400 Robot Manual: Maintenance", and be sure to strictly observe the following items.

⚠ DANGER

- Never incinerate, short-circuit, charge, disassemble, or heat the batteries.
  They could rupture and leak, and if the battery fluid gets on your skin or eyes, it will cause inflammation.
  - If battery fluid gets on your skin, immediately flush with water to wash off.
  - If battery fluid gets in your eyes, immediately flush with water to wash off, and then seek medical attention.
  - Dispose of the batteries in accordance with the customer's regulations.

NOTICE

- Be sure to dispose of any unneeded packing materials in accordance with your company's rules.
  When storing the packing materials, be sure to manage them so that they are not used for purposes other than packing the robot.

This section provides important safety information for disposal of the controller. Read this information together with " STEA1107:TSL3000Controller Manual: Disposal", and be sure to strictly observe the following items.

⚠DANGER

- Never incinerate, short-circuit, charge, disassemble, or heat the batteries. They could rupture and leak, and if the battery fluid gets on your skin or eyes, it will cause inflammation.
  - If battery fluid gets on your skin, immediately flush with water to wash off.
  - If battery fluid gets in your eyes, immediately flush with water to wash off, and then seek medical attention.
  - Dispose of the batteries in accordance with the customer's regulations.

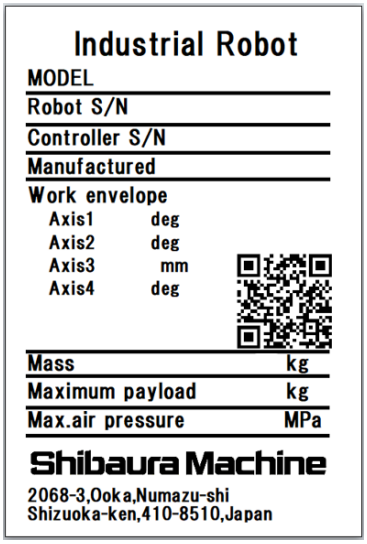
NOTICE

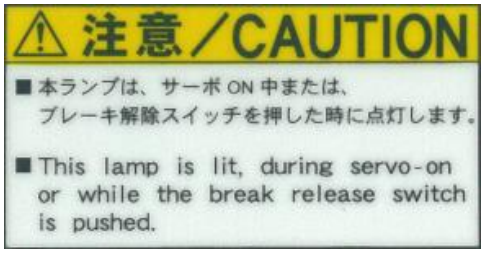
- Be sure to dispose of any unneeded packing materials in accordance with your company's rules. When storing the packing materials, be sure to manage them so that they are not used for purposes other than packing the robot.

1. Warning Labels

1.1 Warning indication contents

The entry contents of the warning labels affixed to the robot and controller are described.

Label layout	Entry contents
	MODEL: Robot model Robot S/N: Robot serial number Controller S/N: Controller serial number Manufactured: Month of manufacture. Year of manufacture Work envelope: Maximum movable range of each joint axis MASS: Robot weight Maximum payload: Maximum payload * QR code: MODEL, SER.NO., and manufacturing instruction number can be read.
• MODEL, SER.NO., and Mfd.DATE are information required for receiving after-sale service. When calling us at the after-sale service agent, notify the service representative of these information. • During installation, transportation, and disposal, be sure to confirm the mass before carrying the robot. Increase the number of persons and use the transport equipment as required. The transport equipment should be operated by an operator who has received training pertaining to the transport equipment.	

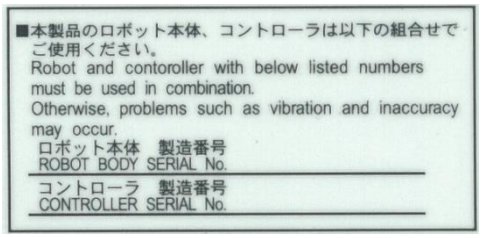
Label layout	Entry contents
	• The lamp on the robot arm 2 is lit during servo ON or when the brake release switch is pressed.
• The lamp on the robot arm 2 is lit during servo ON or when the brake release switch is pressed. If the brake release switch is not pressed, the robot may wait for operation instruction during program operation. Because the robot may start moving suddenly, do not get close to the robot. When getting close to the robot, confirm that the program has stopped, and then switch the operation mode to TEACH. If the brake release switch is pressed, the axis 3 may drop. While using the support that prevents drop, perform the operation.	

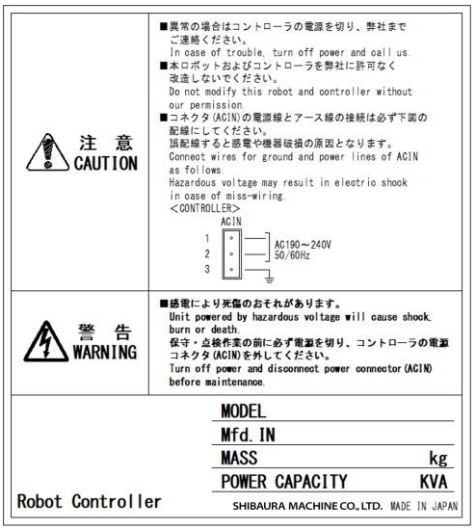
• Label layout	• Description
• 	- Keep your hands away from the operating parts of the robot. Your fingers could get entangled, resulting in an injury.
- If you put your hand near the operating parts of the robot, it could get caught in the robot arm or shaft, resulting in an injury. During transfer for installation, transportation, and disposal, secure the arm and shaft in place using the supplied fixture so that they do not move. • If you must enter the moving range of the robot to perform teaching or maintenance work for unavoidable reasons, the work must be performed by a trained worker. • Install a protective fence to prevent people from entering (entering the moving range of the robot or having a body part get near the operating parts) during operation.	

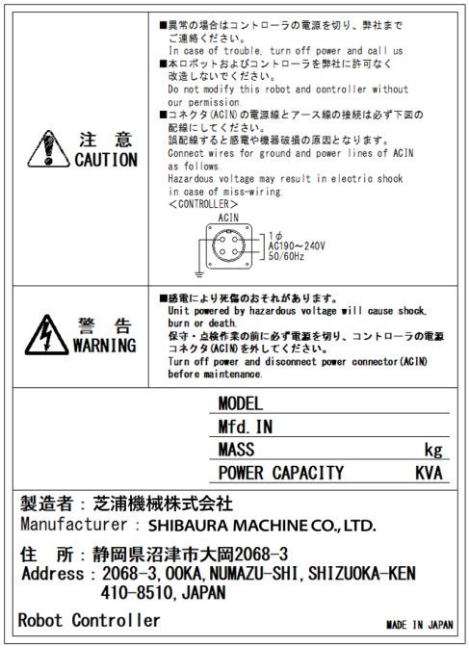
• Label layout	• Description
• 	• - Do not enter into the robot's movable range. Otherwise, an injury could result.
- If you enter the movable range of the robot, you could be killed or injured due to contact with the moving robot. If you must enter the moving range of the robot for unavoidable reasons to perform teaching or maintenance work, this must be performed by a trained operator. Implement measures so that no one other than workers performing teaching or maintenance can enter the movable range.	

• Label layout	• Description
• 	• -High temperature • This is affixed to locations that could become hot when operating the robot. • Touching these locations could result in a burn. • Wait until these high-temperature locations have cooled before performing any maintenance, inspection, or adjustment work.
• Wait until these high-temperature locations are fully cooled before performing any work where the robot is touched. If necessary, protect your hands using thermal-resistant gloves or similar protection when performing the work.	

• Label layout	• Description
• 	• - Be sure to always turn off the controller before performing any maintenance or inspection work. Otherwise, death or serious injury could result due to an electric shock or unexpected movement of the robot.
- If maintenance or inspection work will be performed without turning off the primary power of the controller, the worker could accidentally touch an energized part or charged part, resulting in an electric shock. • Also, the robot could start to move due to an unexpected command issued from an external device, contacting the worker and resulting in death or serious injury. • Before performing any maintenance or inspection work, be sure to always turn off the primary power, and disconnect the power connector. Also, be sure to post signage that work is in progress so that the primary power will not be turned on during the work. If you must turn on the power to perform work for unavoidable reasons, the work must be performed by a trained worker.	

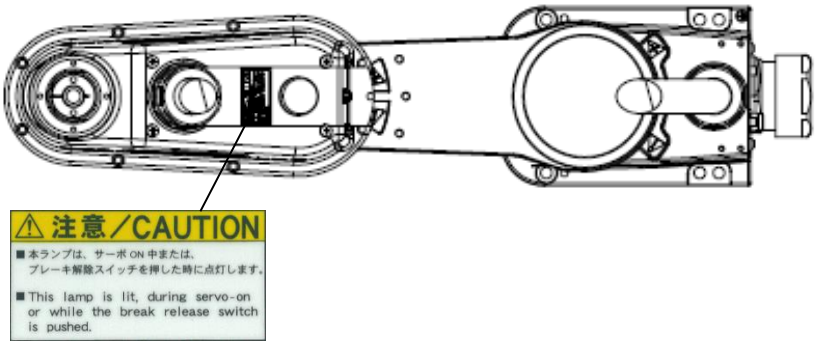
Label layout	Entry contents
	• Use the robot body and controller in combination with the correct serial numbers entered on the label. • Robot body serial number • Controller serial number
• If the robot is operated in combination with wrong serial numbers, a phenomenon such as coordinate shift occurs because the individual parameters of the robot do not match. Confirm the serial numbers of the robot and controller and connect and install them in a correct combination.	

Label layout	Entry contents
	• If an abnormality occurs, turn off the power to the controller, and then contact SHIBAURA MACHINE. • Do not modify this robot and controller without our permission. • Be sure to connect power wire and ground wire of the connector (ACIN) as shown in the following diagram. Mis-wiring causes an electric shock or equipment destruction. • An electric shock may result in death or injury. Before maintenance or inspection, be sure to turn off the power and remove the power connector (ACIN) of the controller • MODEL: Controller model • Mfd.IN: Manufacture date • MASS: Controller mass • POWER CAPACITY: Power capacity
• If the robot and controller are modified without our permission, they may cause failure or runaway. Do not modify them without our permission. For technical inquiry, call us at the after-sale service agent. • Mis-wiring the power wire and ground wire for use causes an electric shock or equipment destruction. Before connecting them, use a tester to confirm whether they are correctly wired. Also, confirm that the primary power supply voltage is within the use range. • If maintenance or inspection is performed without turning off the primary power to the controller, the conductive or charging portion may be touched by mistake, resulting in an electric shock to cause death or injury. When performing maintenance or inspection, be sure to turn off the primary power and remove the power connector. Also, post a notice indicating that operation is in progress to prevent the primary power from being turned on during operation. When having to perform operation with the power turned on, a trained operator should always perform the operation. • MODEL and Mfd.IN are information required for receiving after-sale service. When calling us at the after-sale service agent, notify the service representative of these information. • During installation, transportation, and disposal, be sure to confirm the mass before carrying the robot. Increase the number of persons and use the transport equipment as required. The transport equipment should be operated by an operator who has received training pertaining to the transport equipment.	

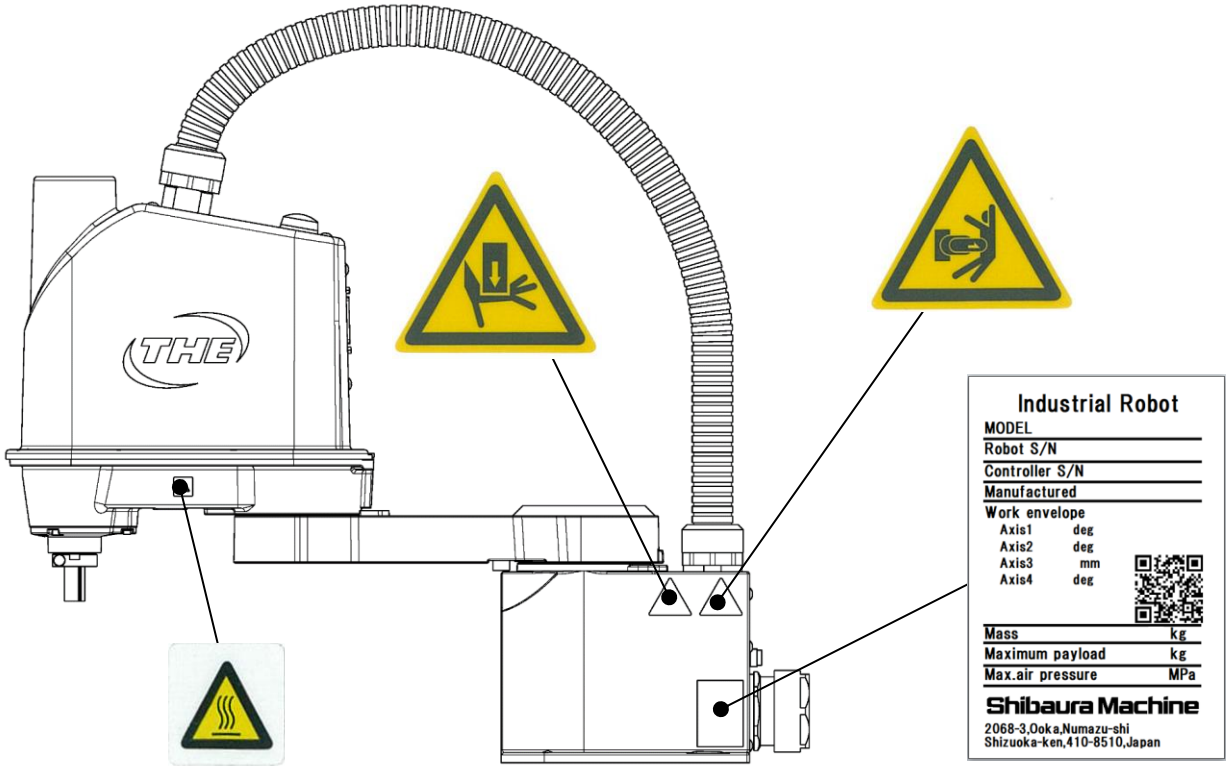
Label layout	Entry contents
	• If an abnormality occurs, turn off the power to the controller and contact SHIBAURA MACHINE. • Do not modify this robot and controller without our permission. • Be sure to connect power wire and ground wire of the connector (ACIN) as shown in the following diagram. Mis-wiring causes an electric shock or equipment destruction. • An electric shock may result in death or injury. Before maintenance or inspection, be sure to turn off the power and remove the power connector (ACIN) of the controller. • MODEL: Controller model • Mfd.IN: Manufacture date • MASS: Controller mass • POWER CAPACITY: Power capacity

- If the robot and controller are modified without our permission, they may cause failure or runaway.
  Do not modify them without our permission. For technical inquiry, call us at the after-sale service agent.
- Mis-wiring the power wire and ground wire for use causes an electric shock or equipment destruction.
  Before connecting them, use a tester to confirm whether they are correctly wired. Also, confirm that the primary power supply voltage is within the use range.
- If maintenance or inspection is performed without turning off the primary power to the controller, the conductive or charging portion may be touched by mistake, resulting an electric shock to cause death or injury.
  When performing maintenance or inspection, be sure to turn off the primary power and remove the power connector. Also, post a notice indicating that operation is in progress to prevent the primary power from being turned on during operation.
  When having to perform operation with the power turned on, a trained operator should always perform the operation.
- MODEL and Mfd.IN are information required for receiving after-sale service. When calling us at the after-sale service agent, notify the service representative of these information.
- During installation, transportation, and disposal, be sure to confirm the mass before carrying the robot.
  Increase the number of persons and use the transport equipment as required. The transport equipment should be operated by an operator who has received training pertaining to the transport equipment.

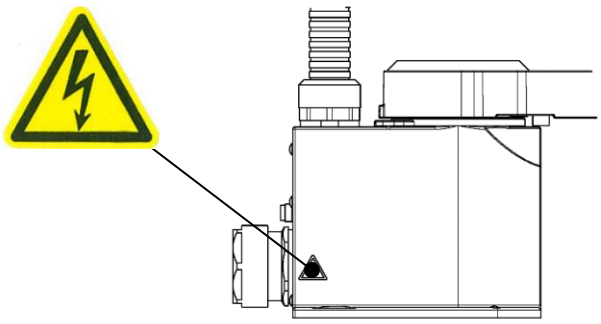
1.2 Locations of robot warning labels (for THE400)



Top view of robot

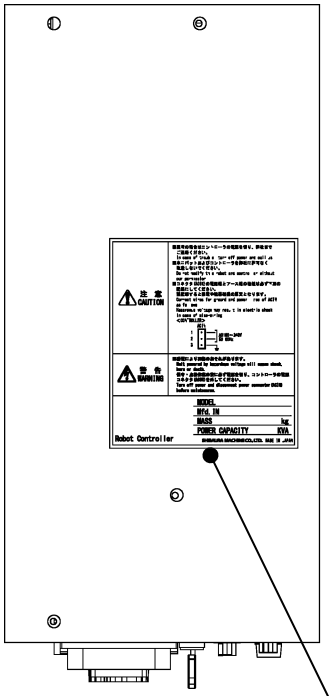


Right side view of robot

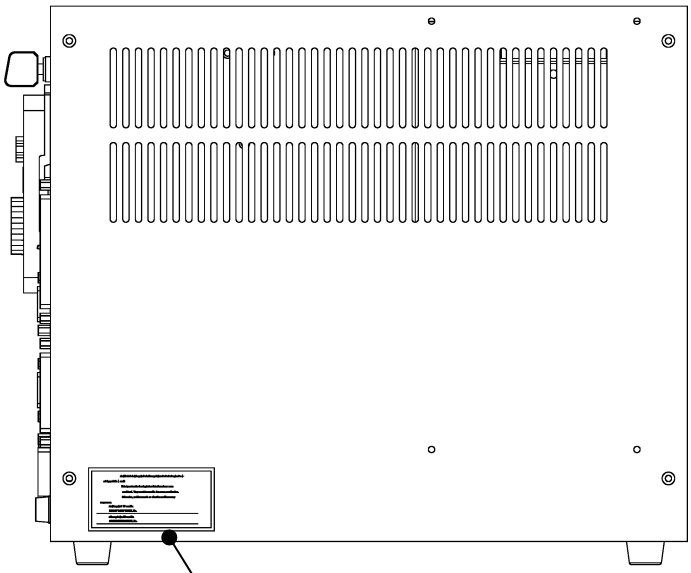


Left side view of robot

1.3 Locations of controller warning labels (for TSL3000)

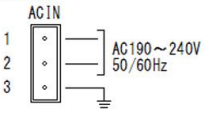


Top view of controller



Right side view of controller

■異常の場合はコントローラの電源を切り、弊社までご連絡ください。
In case of trouble, turn off power and call us.
■本ロボットおよびコントローラを弊社に許可なく改造しないでください。
Do not modify this robot and controller without our permission.
■コネクタ(ACIN)の電源線とアース線の接続は必ず下図の配線にしてください。
誤配線すると感電や機器破損の原因となります。
Connect wires for ground and power lines of ACIN as follows.
Hazardous voltage may result in electric shock in case of miss-wiring.
<CONTROLLER>



注 意
CAUTION

■感電により死傷のおそれがあります。
Unit powered by hazardous voltage will cause shock, burn or death.
保守・点検作業の前に必ず電源を切り、コントローラの電源コネクタ(ACIN)を外してください。
Turn off power and disconnect power connector (ACIN) before maintenance.

警 告
WARNING

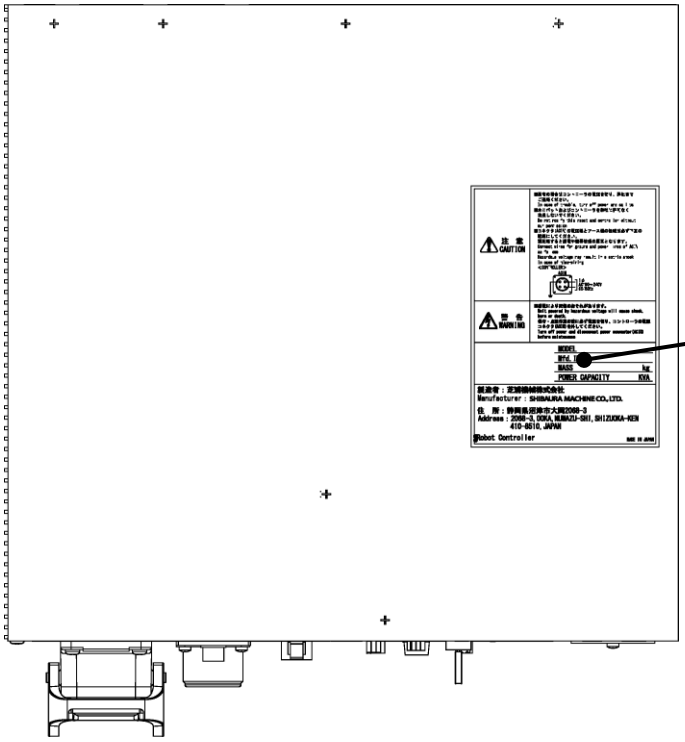
MODEL
Mfd. IN
MASS kg
POWER CAPACITY KVA

Robot Controller

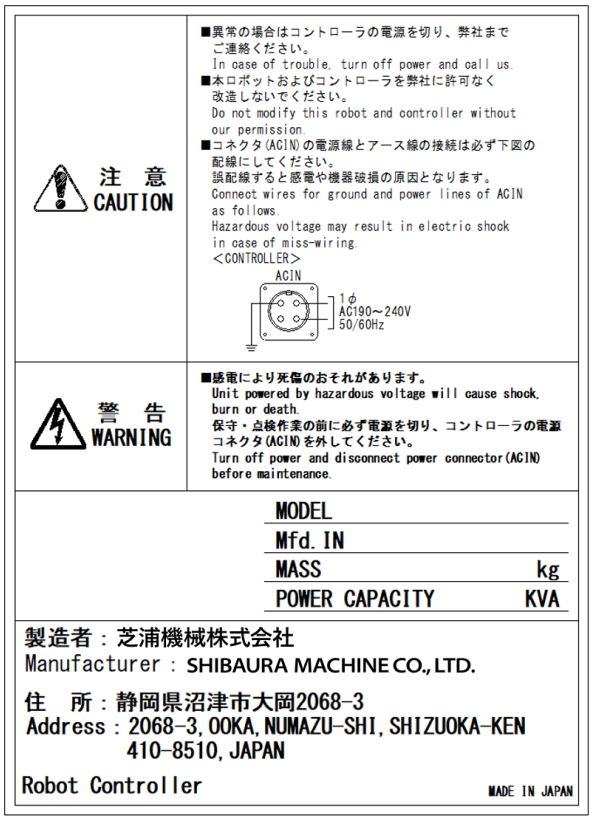
SHIBAURA MACHINE CO., LTD. MADE IN JAPAN

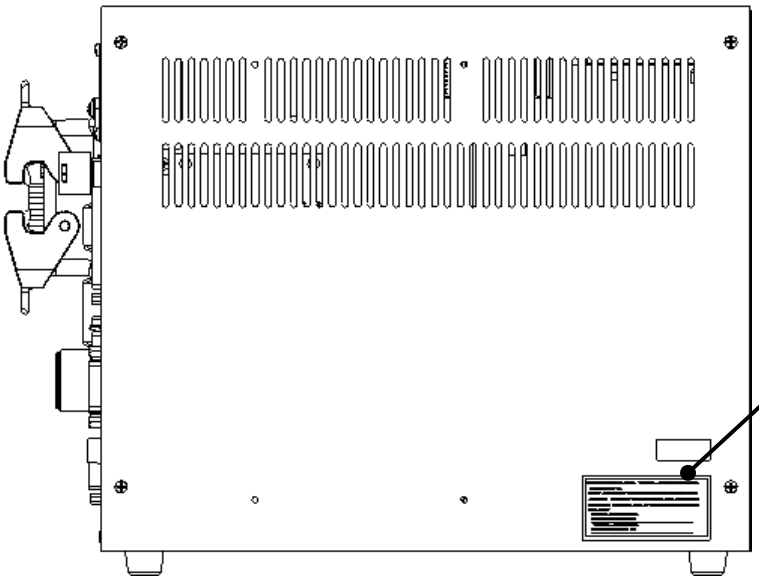
■本製品のロボット本体、コントローラは以下の組合せでご使用ください。
Robot and contoroller with below listed numbers must be used in combination.
Otherwise, problems such as vibration and inaccuracy may occur.
ロボット本体 製造番号
ROBOT BODY SERIAL No.
コントローラ 製造番号
CONTROLLER SERIAL No.

1.4 Locations of controller warning labels (for TSL3000E)



Top view of controller





Right side view of controller



2. Safety Measures

This section describes the necessity of safety measures, safety functions provided for the robot and robot controller, and general safety measures to be taken.

2.1 Cautions on safety

Our SCARA robot and robot controller are equipped with various safety functions.
When the robot is actually operated, however, the following dangers will be supposed.

a) Danger supposed in normal automatic operation

- Operator's mis-operation and mis-judgment, and incomplete program.
- Unexpected robot movement and release or drop of workpiece due to fault of an electronic control device.

b) Danger supposed at teaching and inspection

- Danger of an operator entering the movable range of the robot.
- Operator's negligence from confusion or experience at generation of an unexpected abnormality, and operator's mis-operation due to shortage of behavior and knowledge.
- Approach of an operator to the robot due to unexpected complex movement of the robot.
- Abnormal movement, etc. caused by mis-wiring, poor contact, deterioration and noise

c) Danger supposed in a related machine, etc.

- Sudden movement of the robot with a command from a related machine, etc.
- Sudden movement of a related machine after the robot movement.
- Danger of an operator being caught or entangled in the robot when teaching, inspecting or adjusting the robot while moving a related machine.

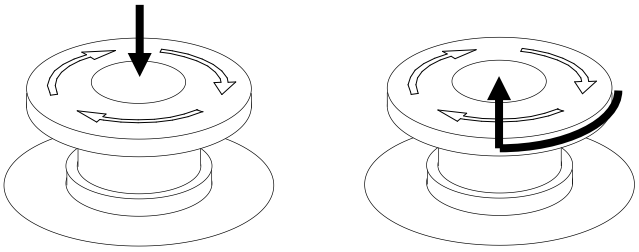
To use the robot safely, safety measures should be taken according to the operating conditions.
Otherwise, an unexpected disaster may occur.
If there are safety rules and regulations, strictly observe the rules and regulations in each country. Also, refer to all manuals relating to this robot and robot controller.

2.2 Safety functions

This robot and robot controller are equipped with various safety functions and additional protection functions.

2.2.1 Emergency stop function

An emergency stop pushbutton switch is provided on the teach pendant. This switch can stop the robot immediately and shut down the robot's power source. (Stop category 1) The emergency stop function is always enabled regardless of the operating condition of the robot.
The emergency stop pushbutton switch is equipped with a lock mechanism. To reset the emergency stop state, pull up the upper part of the switch while turning it as shown in the following figure.
The controller also has the external emergency stop contacts, so an emergency stop switch can be added as required. Select ISO 13850-compliant emergency stop switches as the additional ones.



Emergency stop switch ON　　Emergency stop switch OFF

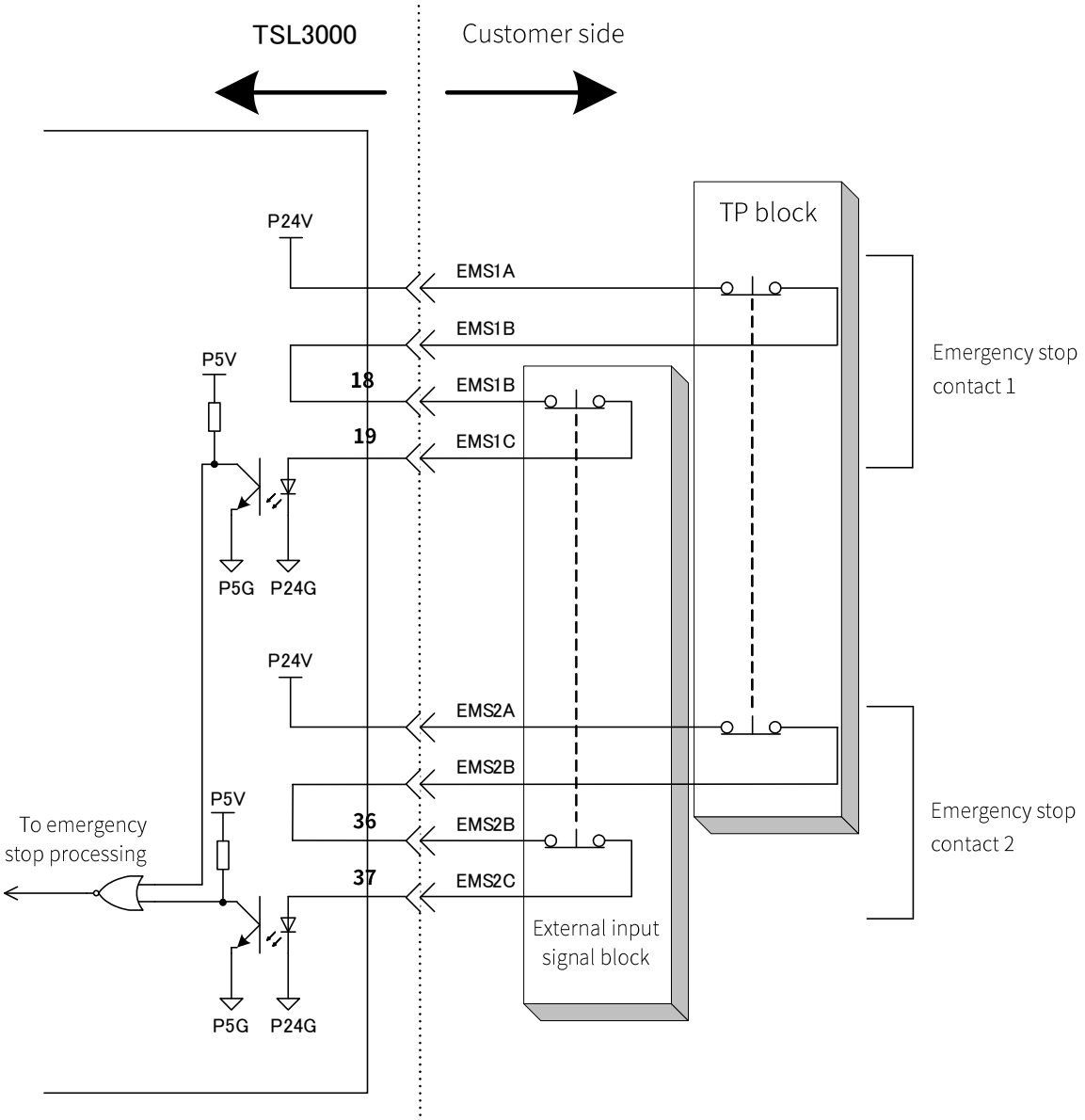
Precautions for the emergency stop switch
Note the following to select an emergency stop pushbutton switch.

- An emergency stop device must be installed at a location that requires emergency stop and is easy to approach.
- An emergency stop device must be a device that, after functioning, does not allows the circuit to automatically recover until restored manually, and that has a structure where the contact does not recover until the device is reset.
- An emergency stop device must has an appropriate shape like those listed below so that it can eliminate danger according to the structure and characteristics of the machine.
    1) Mushroom-shaped pushbutton
    2) Cable pull switch, lever switch
    3) Switch operated by belly or knee
    4) Pedant switch with no protection cover
- The actuator of an emergency stop device with a pushbutton switch must be red, and the background color of the surrounding must be yellow.
- The cable of an emergency stop device with a cable pull switch must always have a proper tension, and must be distinguished by color (red and yellow).

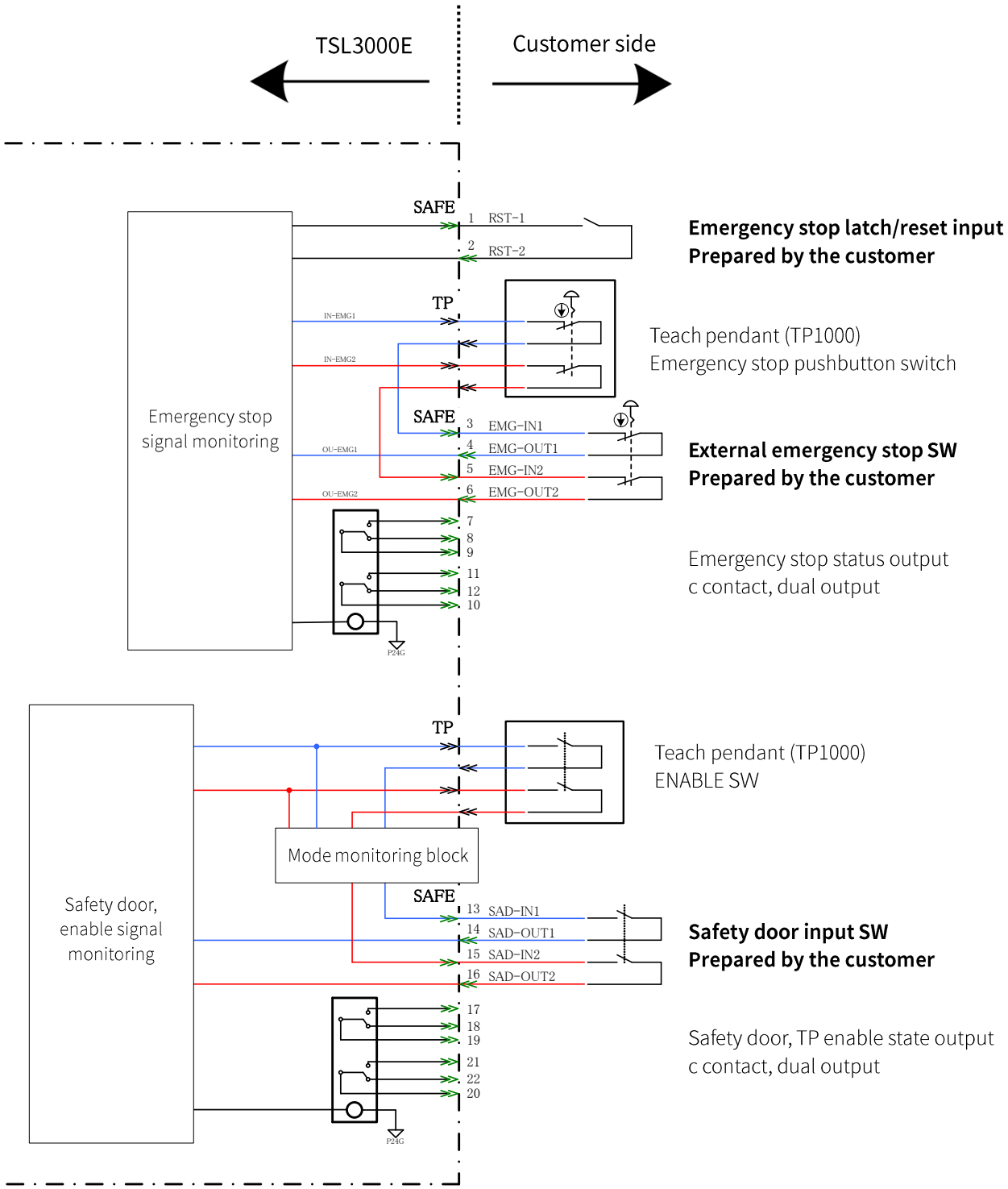
2.2.2 Safety contact input function

TSL3000 and TSL3000E have the duplicated safety contact input function that is a different system from emergency stop. Connect interlock switches, such as a limit switch and a foot switch, which operate simultaneously with a door of a safety fence.

1) External emergency stop contacts of TSL3000 are shown in the figure below.
Connect emergency stop button switches to the 18-19 pins and 36-37 pins of the INPUT connector on the front of TSL3000 for use. For output common Type-N and Type-P, the emergency stop button switches are also connected to the same pin numbers.



2) External emergency stop contacts of TSL3000E are shown in the figure below.
Connect emergency stop button switches to the 3-4 pins and the 5-6 pins of the SAFE connector on the front of TSL3000E, and also fit the safety door to the 13-14 pins and 15-16 pins and connect it to interlock switches, such as a limit switch and a foot switch, which operate simultaneously with a door of a safety fence.

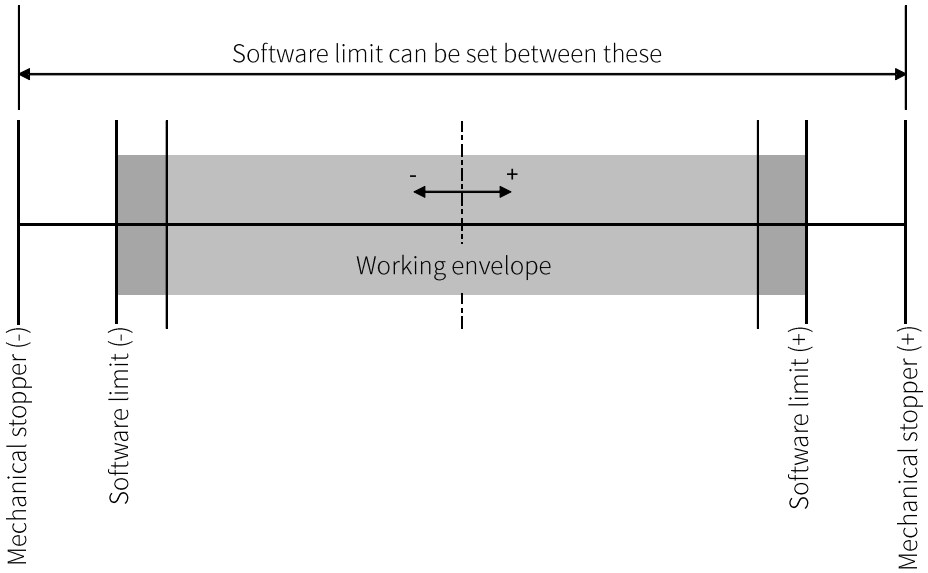


2.2.3 Working range limiting function

This robot is equipped with mechanical stoppers to prevent overrun.
The mechanical stoppers are provided on axis 1, axis 2, and axis 3 in order to prevent the movable part of the robot from overrunning.
The robot working range can be limited by changing the mechanical stopper positions, depending on the robot model.
For change of the mechanical stopper positions, refer to the Instruction Manual “THE Series Robot Manual: Installation.”

Software limits can be set as the controller side functions. The software limits are the auxiliary functions of the mechanical stoppers that prevent collision with the mechanical stoppers due to a mis-operation during manual guide or program error. The software limits should be set inside the robot working range limited by the mechanical stoppers.
For change of software limits, refer to the Instruction Manual “THE Series Robot Manual: Installation.”

When the robot operates abnormally, it is unlikely that the software limits are able to limit the working range.
The safety functions aimed to protect human body should be performed with the mechanical stoppers.



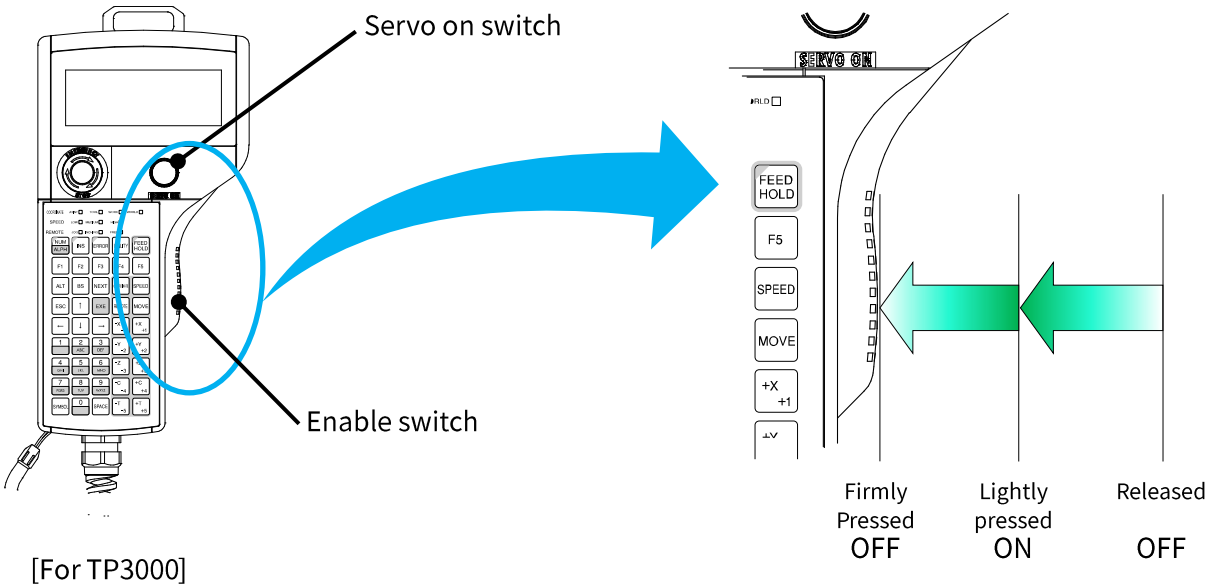
2.2.4 Operation switch on teach pendant

An enable switch is provided on the teach pendant in order to enhance the safety. When guiding the robot in manual mode, the operator holds the teach pendant and keeps the enable switch in “gently pressed” state to reset the emergency stop state and turn on the servo power.

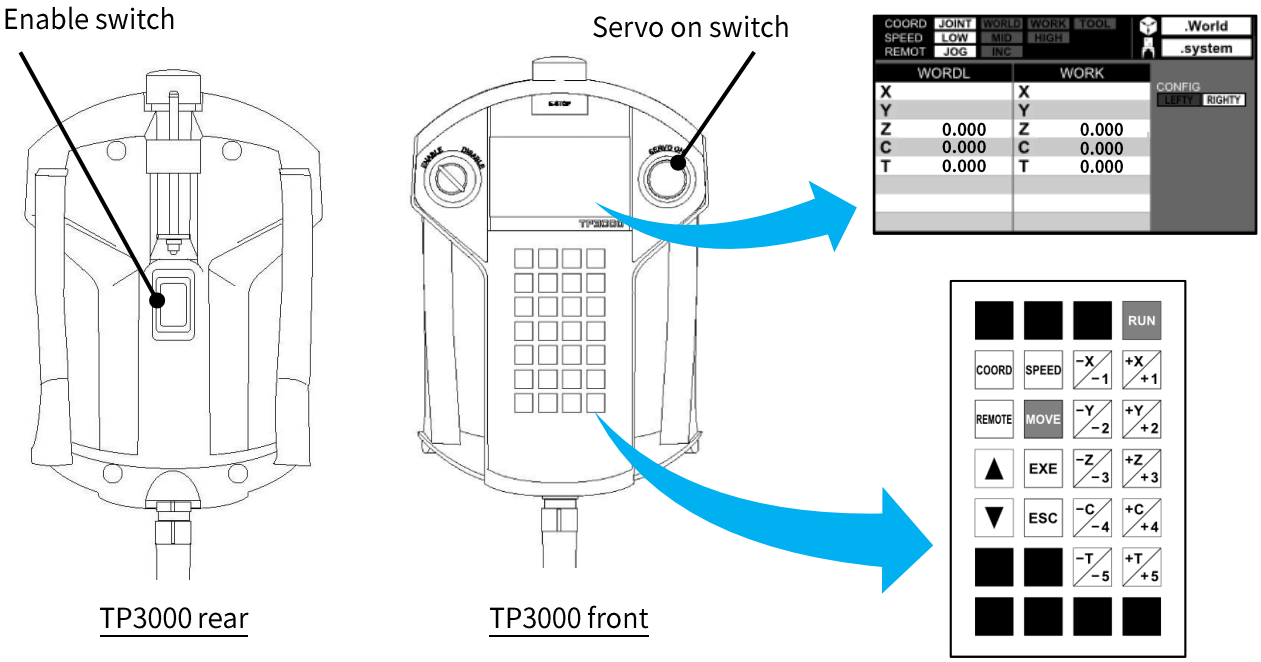
The enable switch is of the three-position type; the switch stops the robot immediately and turns off the robot’s power source when it is “released” or “firmly pressed.”

The servo cannot be turned on when the enable switch is OFF. Keep the enable switch ON when guiding the robot with the teach pendant.

[For TP1000]



[For TP3000]



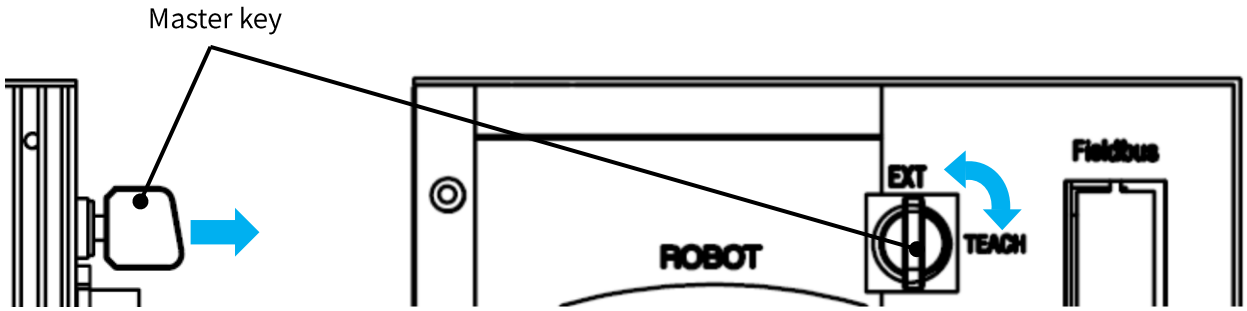
TP3000 rear

TP3000 front

When the enable switch is “gently pressed,” the switch turns ON and the screen and key sheet change to the JOG mode. When the switch is “released” or “firmly pressed,” they do not change to the JOG mode.

2.2.5 Master key switch in front of controller

The master key switch is provided to select the TEACH or EXT mode of the control panel. When having to enter the working range in order to do teaching operation or inspection, the operator should select the TEACH mode and carry a key during the operation to prevent another operator from changing the mode.



Side view of master key switch

Front view of master key switch

2.2.6 CE-compliant safety function (TSL3000E)

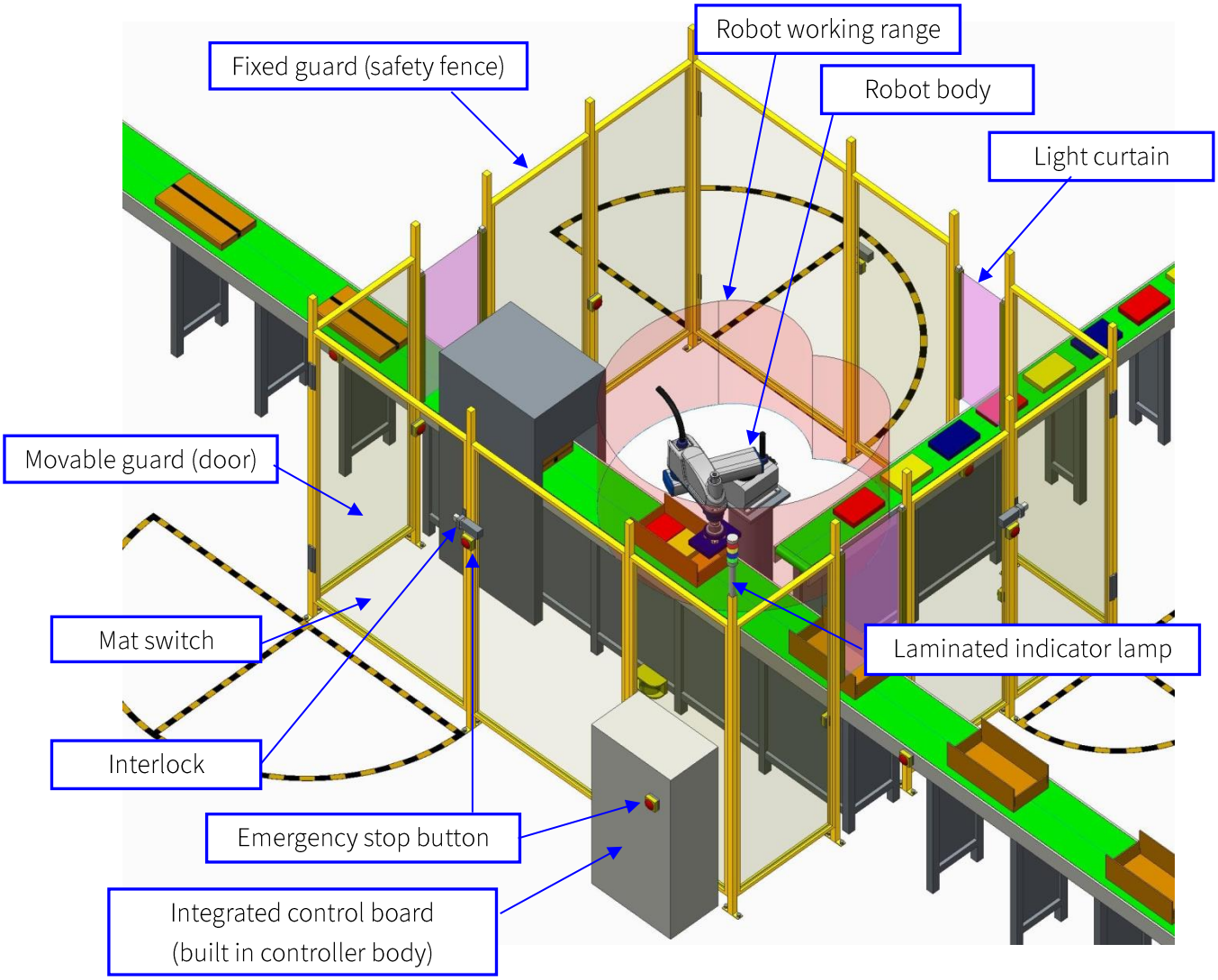
TSL3000E refers to TSL3000 that is compliant with safety category 3. For the difference from TSL3000, see the explanation in "3. Applicable Standards and Safety Performance."

The TSL3000E controller is equipped with the input terminal to which the safety circuit is connected.

2.3 Examples of safety measures

Most of disasters caused by the robot originate from unsafe human behavior. When using the robot, the operator should foresee what will involve danger and try to prevent such a dangerous condition. Operation should be done only after all safety conditions are satisfied. Examples of main safety measures are shown below.

2.3.1 General cautions on using the robot



1) When installing the robot, provide a space required for performing the operation during installation and subsequent operations (teaching, operation, maintenance and disposal) safely.
   - ISO 13854: Minimum gaps to avoid crushing of parts of the human body

2) Identify the dangerous area, and install safety fences, etc. to prevent entry of any person. The dangerous area signifies an area near the robot, where a person who has entered it will jeopardize.
   - ISO14120: General requirements for the design and construction of fixed and movable guards
   - ISO13857: Safety distances to prevent hazard zone being reached by upper and lower limbs

3) Install an emergency stop switch at a location where it can be pressed immediately by an operator who has determined the occurrence of abnormality outside and inside the safety fence, etc., and connect it to the external emergency stop contacts (EMS1B to 1C and EMS2B to 2C) of the controller.
   - ISO13850: Emergency stop

4) Install interlock devices such as interlock switch, light curtain and mat switch at the entrance and exit doors of the safety fence and connect them to the safety contact input of the controller so that the robot stops if a person enters the dangerous area. The interlock devices should be electrically independent break-contact (normally-closed contact) devices.
   - ISO14119: Interlocking devices associated with guards
   - IEC61496-1,-2: Electro-sensitive protective equipment
   - ISO13855: Positioning of safeguards with respect to the approach speeds of parts of the human body
   - ISO13856-1: Pressure-sensitive protective devices

5) Install the controller at a safe location where an operator can overlook the operation of the robot.

6) To perform transportation, installation, teaching (teaching, program and operation check), maintenance (inspection, adjustment and repair) and disposal, be sure to observe the laws, rules and regulations defined in each country. Also, display the current condition of the robot to prevent an unrelated person from carelessly entering the working range and operating the robot.

7) Before daily operation, perform the following check.
   <Check before operation>
   - Visually confirm that the robot body, controller, peripheral equipment and cables are in good condition.
   - Make sure that no object stands in or new the working range of the robot and peripheral equipment.
   - Make sure that the emergency stop and other safety devices operate properly.
   - Make sure that no abnormal noise or vibration is involved in the robot operation.

8) Be sure to operate the pushbutton switches provided on the teach pendant while visually confirming them.

2.3.2 Cautions on operation

In principle, perform teaching and inspection outside the dangerous area of the robot. To perform them inevitably, strictly observe the following cautions.

1) Teaching should be performed after the drive source is shut off (servo OFF) whenever possible. The drive source should not be allowed to enter automatic state or to be recovered (servo ON) by other operator.
   Inspection should be performed after the primary power is shut off whenever possible. The primary power should not be allowed to enter automatic state or to be recovered by other operator.

2) The switches that change the operating state of the robot (start switch, stop switch, etc.) should not be allowed to be operated by a person other than the operator (by posting the display board, locking the operator panel, arranging watcher, etc.).

3) The following work regulations should be established so that the operation is performed in accordance with the regulations.
   ① Operation methods and procedures for starting the robot, handling the switches, teaching, inspection, and confirming them
   ② Operating speed suitable for situations (the speed is desirably made as slow as possible.)
   ③ Method for exchanging signals between operators
   ④ Recovery method for an abnormality
   ⑤ Method for resetting abnormal state and confirming safety before restarting the robot that has stopped abnormally
   ⑥ Other following methods for preventing the danger caused by unexpected movement of the robot and mis-operation
      • Methods for limiting operation of switches (start switch, stop switch, etc.) that select the operating state of the robot to an operator (by posting the display board, locking the operator panel, arranging watcher, etc.)
      • Method for allowing an operator in the dangerous area to hold the emergency stop device (by using the teach pendant)
      • Position and posture in which safety of an operator can be secured
      • Types and determination method of abnormalities (method for confirming alarm, etc.)
      • Method for preventing and reducing noise
      • Method for changing setup and confirming it (program change procedure, teaching data to be change, method for confirming it, etc.)
      • Method for exchanging signals between robot operator and related equipment operator

   * The work regulation should try to be prepared by inviting suggestions about the description suitable for respective sites from concerned operators, facility design engineers, etc.

4) The operations should always be performed by two persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other.

5) The operator should do the job in an attitude ready to press the emergency stop pushbutton switch at any time. Also, the operator should perform the job at a position from which the operator can evacuate immediately at the time of an emergency after confirming the robot working range and shields in the surroundings. Also, the operator should be careful so as not to turn operator's back to the robot during operation.

6) The supervisor should keep on the job at a position where the supervisor can overlook the entire robot system and operate the emergency stop pushbutton switch at the time of an emergency. Also, the supervisor should keep anyone from entering the dangerous area.

2.3.3 Other cautions

1) The gripping unit of the robot should not stick out, except for the part required for operation. Also, even at a sudden stop due to power failure, malfunction or emergency stop during operation, the robot should hold a workpiece in a stable posture.

2) If the robot has malfunctions, turn off the power, remove the cause of the malfunction, maintain the peripheral equipment and completely restore the malfunctioned robot. Then, confirm its movement a low speed. Even if the robot has stopped, do not approach the dangerous area immediately.

3) Turn off the power when the robot is not in use.

Note) Because of space limitations, this document covers only important safety precautions for the robot, but not all general safety information. Before carrying out work, read safety documents compliant with the laws, regulations and standards in each country or receive instructions from the consultants who are familiar with the rules, regulations and standards.
Also, carry out risk assessment and take measures to reduce risk.

2.3.4 Recommended protector

Observe the standards, regulations and rules in each country to carry out risk assessment, and then consider and select use.

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	
Earplug	Protection part: Ears Use: Protect from noise*1.	

*1): For details about the noise level and the method for measuring it, refer to the Instruction Manual "THE Series Robot Manual: Specifications." SHIBAURA MACHINE recommends that the protectors be worn for the operation in the environment where a single robot produces generates noise at 85 db or more based on the Guideline for the Prevention of Noise-Induced Impairments (Labor Standards Bureau Issue No. 546) in accordance with the laws in Japan.

3. Applicable Standards and Safety Performance

1) Combination of robot and controller and applicable standards

Robot	Controller	Machinery directive 2006/42/EC	Low voltage directive 2006/95/EC	EMC directive 2004/108/EC
THE400	TSL3000E	ISO10218-1:2011 ISO13849-1:2008	IEC60204-1:2006 IEC61800-5-1:2007	EN55011:2009 EN61000-6-2:2005

2) Combination of robot and controller and safety performance

Robot	Controller	ISO13849-1:1999 Safety category	ISO13849-1:2008 Performance level
THE400	TSL3000E	Category 3	PL=d

The EU Directive with which SHIBAURA MACHINE robot connected with TSL3000E is compliant to have CE marking includes machinery directive (2006/42/EC), low-voltage directive (2006/95/EC), and EMC directive (2004/108/EC). Also, this robot makes self-declaration of CE marking as a semi-finished machine. Completing the robot as a robot system requires that validity of the robot be confirmed as a result of properly carrying out risk assessment based on EU Directive in a combination with the end effector and auxiliary safety devices (e.g., safety fences, enclosure, etc.).